MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, LOW NOISE OPERATIONAL AMPLIFIERS MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low noise operational amplifiers. Three product assurance classes and a choice of case outline and lead finish are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Single operational amplifier, low noise, undercompensated
02	Dual operational amplifier, low noise, compensated

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510 Appendix C)
G	A-1 (8 lead can)
P	D-4 (8 lead DIP, 1/4" x 3/8")

1.3 Absolute maximum ratings.

Supply voltage (V_{CC})	±22 V
Input voltage (V_{IN})	±V_{CC}
Differential input voltage range	±.5 V
Output short circuit duration	1/
Lead temperature (soldering, 60 seconds)	300°C
Storage temperature	-65°C to +150°C
Junction temperature	T_J = 175°C

1.4 Recommended operating conditions.

Supply voltage range	±3.0 to ±20.0 volts
Ambient operating temperature range	-55°C to +125°C

1/ Output may be shorted to ground indefinitely at V_S = ±15 volts, T_A = 25°C. Temperature and/or supply voltages must be limited to ensure dissipation rating is not exceeded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be adressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation 1/	Maximum θ_{JC}	Maximum θ_{JA}
8-lead can	G	330 mW @ T_A = 125°C	40°C/W	150°C/W
8-lead dual-in-line package	P	400 mW @ T_A = 125°C	35°C/W	120°C/W

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Schematic circuits. The schematic circuits shall be as specified on figure 2.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

1/ All leads welded or soldered to P.C. board.

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TABLE I. Electrical performance characteristics for device types 01, 02.

Characteristic	Symbol	Conditions: $\pm V_{CC}$ = ±15, paragraph 3.4 and figure 7 unless otherwise specified	Limits		Unit
			Min	Max	
Input offset voltage	V_{IO}	1/ T_A = 25°C	-2	2	mV
		1/ -55°C ≤T_A ≤+125°C	-3	3	mV
Input offset voltage temperature sensitivity	dV_{IO}/dt		-8	8	µV/°C
Input bias current	$+I_{IB}$	1/ T_A = 25°C	-800	800	nA
		1/ -55°C ≤T_A ≤+125°C	-1500	1500	nA
	$-I_{IB}$	1/ T_A = 25°C	-800	800	nA
		1/ -55°C ≤T_A ≤+125°C	-1500	1500	nA
Input offset current	I_{IO}	1/ T_A = 25°C	-200	200	nA
		1/ -55°C ≤T_A ≤+125°C	-500	500	nA
Power supply rejection ratio	+PSRR	$+V_{CC}$ = 20 to 10 V, $-V_{CC}$ = -15 V	85	--	dB
	-PSRR	$+V_{CC}$ = 15 V, $-V_{CC}$ = -20 to -10 V	85	--	dB
Common mode rejection ratio	CMRR	V_{CM} = ±12 V	80	--	dB
Adjustment for input offset (01 only)	V_{IO} Adj(+)		4	--	mV
	V_{IO} Adj(-)		--	-4	mV
Output short circuit current	$I_{OS(+)}$	4/ t ≤25 msec	-70	--	mA
	$I_{OS(-)}$	4/ t ≤25 msec	--	70	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01, 02 - Continued.

Characteristic	Symbol	Conditions: $\pm V_{CC}$ = ±15, paragraph 3.4 and figure 7 unless otherwise specified		Limits Min	Limits Max	Unit
Supply current	I_{CC}	T_A = 25°C	01	--	6.5	mA
			02	--	11.0	mA
		-55°C ≤T_A ≤+125°C	01	--	9.0	mA
			02	--	13.0	mA
Output voltage swing (maximum)	V_{OP}	R_L = 600 ohms (01 only)		-10	10	V
		R_L = 2000 ohms (01, 02)		-12	12	V
Open loop voltage gain (single ended)	$A_{VS(\pm)}$	2/ T_A = 25°C		50	--	V/mV
		2/ -55°C ≤T_A ≤+125°C		25	--	V/mV
Open loop voltage gain (single ended)	A_{VS}	3/ T_A = 25°C		50	--	V/mV
		3/ -55°C ≤T_A ≤+125°C		25	--	V/mV
Slew rate	SR(+) and SR(-)	V_{IN} = 10 volts		10	--	V/µs
Settling time	ts(+) and ts(-)	T_A = 25°C, A_V = -1 (see fig. 9) R_L = 600 ohms, C_L = 100 pF 0.1% error		--	4	µs
Transient response (rise time)	$T_{R(tr)}$	A_V = +1 (see fig. 8) R_L = 600 ohms, C_L = 100 pF		--	40	ns
Transient response (overshoot)	$T_{R(os)}$	A_V = +1 (see fig. 8) R_L = 600 ohms, C_L = 100 pF		--	40	%
Input noise voltage density	EN	f_o = 30 Hz (see fig. 10)		--	11	$nV/\sqrt{Hz}$
		f_o = 100 Hz		--	9	$nV/\sqrt{Hz}$
		f_o = 1 kHz		--	5.5	$nV/\sqrt{Hz}$
		f_o = 10 kHz		--	5.0	$nV/\sqrt{Hz}$

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01, 02- Continued.

Characteristic	Symbol	Conditions: $\pm V_{CC} = \pm 15$, paragraph 3.4 and figure 7 unless otherwise specified	Limits		Unit
			Min	Max	
Input noise current density	IN	f_o = 30 Hz (see fig. 10)	--	7	pA/ Hz
		f_o = 100 Hz	--	5	pA/ Hz
		f_o = 1 kHz	--	2	pA/ Hz
		f_o = 10 kHz	--	2	nV/ Hz
Broadband input noise voltage	$N_{I(BB)}$	f = 10 Hz - 10 kHz R_S = 50 ohms (see figure 11)	--	500	nV rms
Channel separation	C_S	Device type 02 only (see figure 12)	100	--	dB

1/ Tested at a) $V_{CM} = 0$, $V_{CC} = \pm 5$ V, b) $V_{CM} = \pm 12$, $V_{CC} = \pm 15$, c) $V_{CM} = 0$, $V_{CC} = \pm 15$.

2/ V_{OUT} = 0 to +10 for $A_{VS(+)}$ and V_{OUT} = 0 to -10 for $A_{VS(-)}$. R_L = 600 ohms (01 only); R_L = ZK (02 only).

3/ Tested at a) $V_{CC} = \pm 5$ V, $V_{OUT} = \pm 2$ V, R_L = 600 ohms (01 only), b) $V_{CC} = \pm 5$ V, $V_{OUT} = \pm 2$ V, R_L = 2000 ohms (01, 02).

4/ Continuous short circuit limits are considerably less than the indicated test limits since maximum power dissipation cannot be exceeded.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range for supply voltages from ±3 to ±20 vdc. Unless otherwise specified, source resistance (Rs) shall be 50 ohms for all tests. For dual packages the idle device shall be connected as a ground follower.

3.4.1 Offset null circuits. The nulling inputs shall be capable of being nulled 1 mV beyond the specified offset voltage limits for -55°C $\leq T_A \leq$ 125°C using the circuit of figure 3.

3.4.2 Instability oscillations. The devices shall be free of oscillations when operated in the test circuits of this specification.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 49 (see MIL-M-38510, Appendix E).

TABLE II. Electrical test requirements.

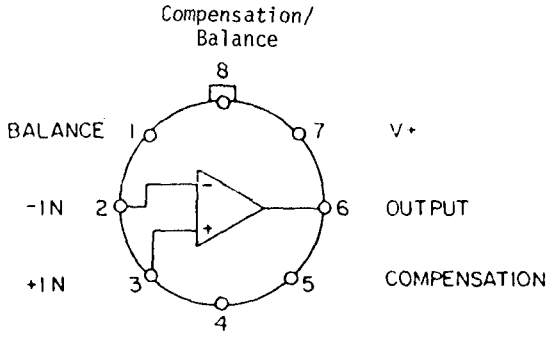
MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,4	1*,2,3,4	1
Group A test requirements (method 5005)	1,2,3,4,5,6,7, 8,12**	1,2,3,4,5,6,7	1,2,3,4,7
Group C end-point and group B class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1,2,3	1	1
Additional electrical subgroups for group C periodic inspection	N/A	8,12	5,6,8,12

* PDA applies to subgroup 1 (see 4.3d).
** See 4.4.1c.

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Device Type 01

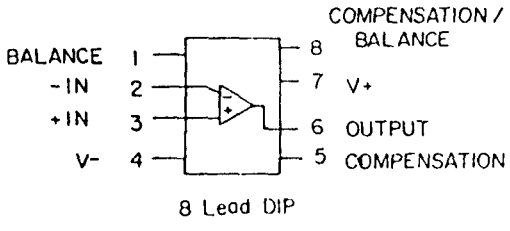
Case G



8 Lead can

Device Type 01

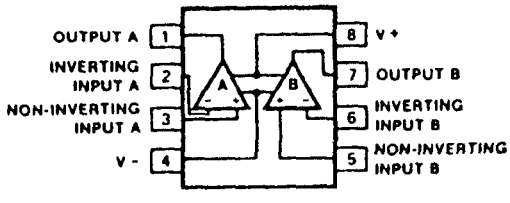
Case P



8 Lead DIP

Device Type 02

Case P

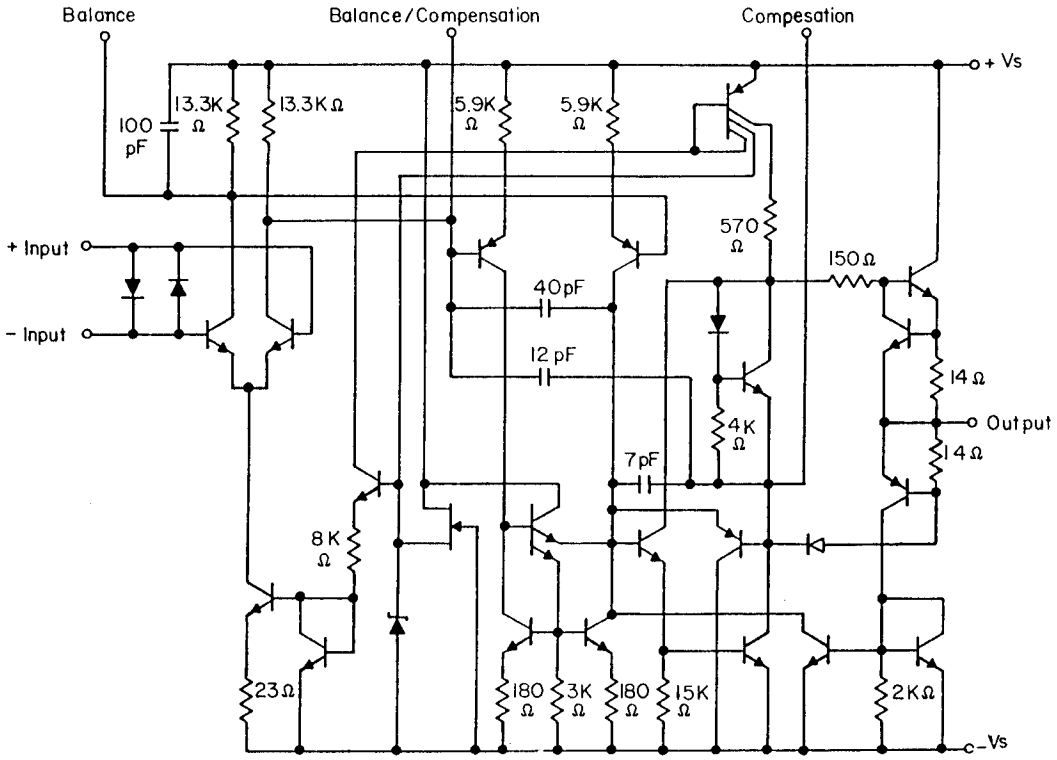


8 Lead DIP

FIGURE 1. Terminal connections and logic diagram.

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Device types 01 and 02



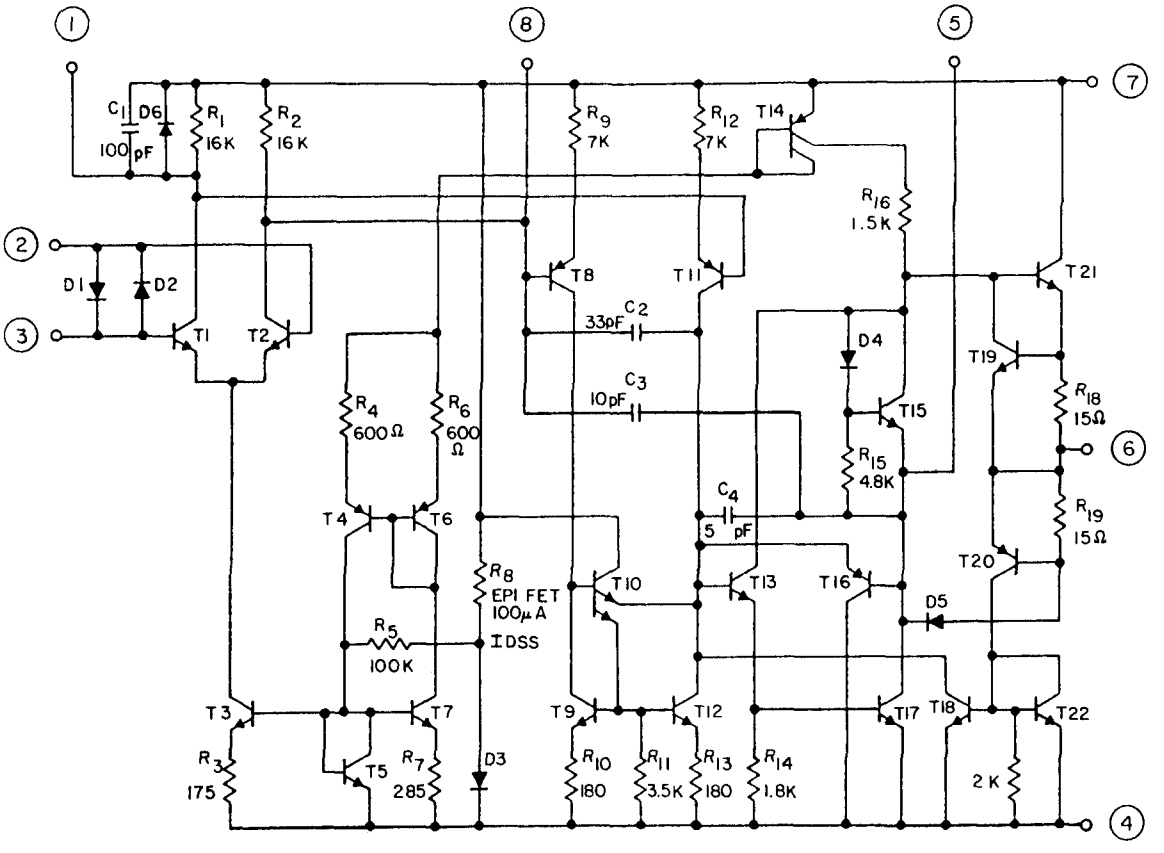
(Circuit shown is 1/2 device type 02)

RAYTHEON

FIGURE 2. Schematic circuit.

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Device Types 01



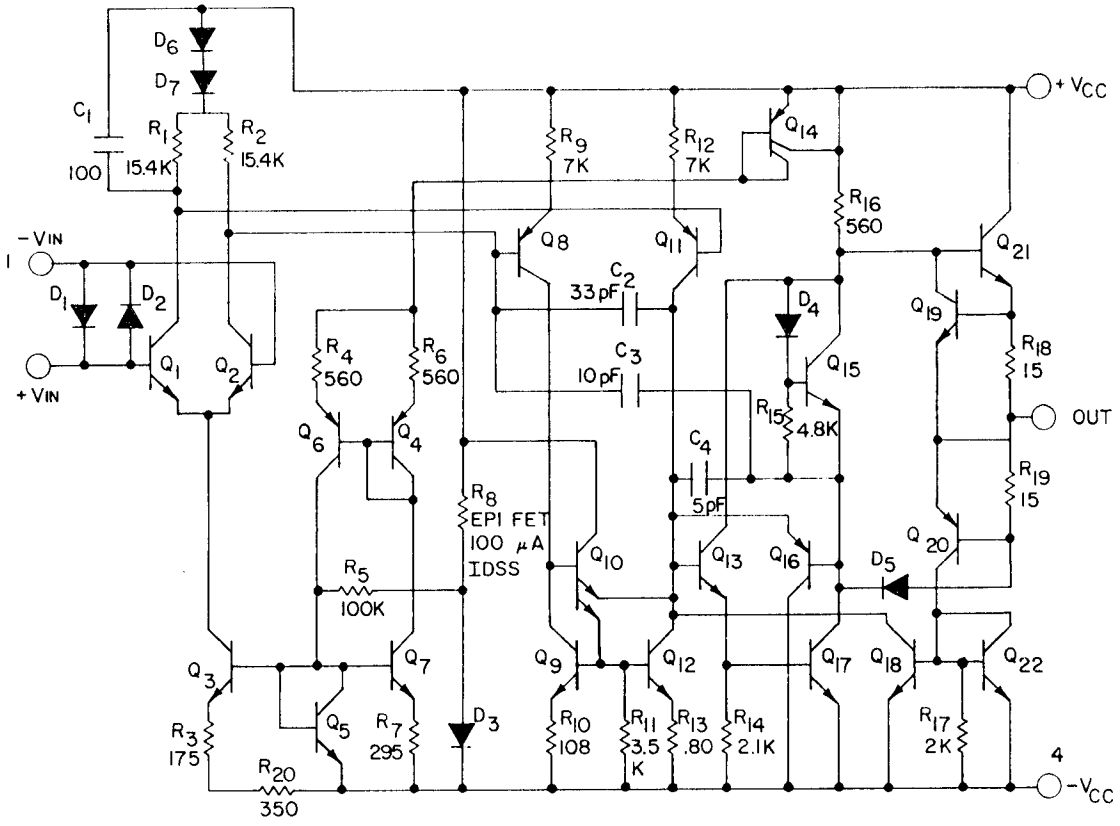
SIGNETICS

NOTE: Resistance values are in ohms.

FIGURE 2. Schematic circuit

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Device type 02



(Circuit shown is 1/2 device type 02)

SIGNETICS

FIGURE 2. Schematic circuit.

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Device Type 01

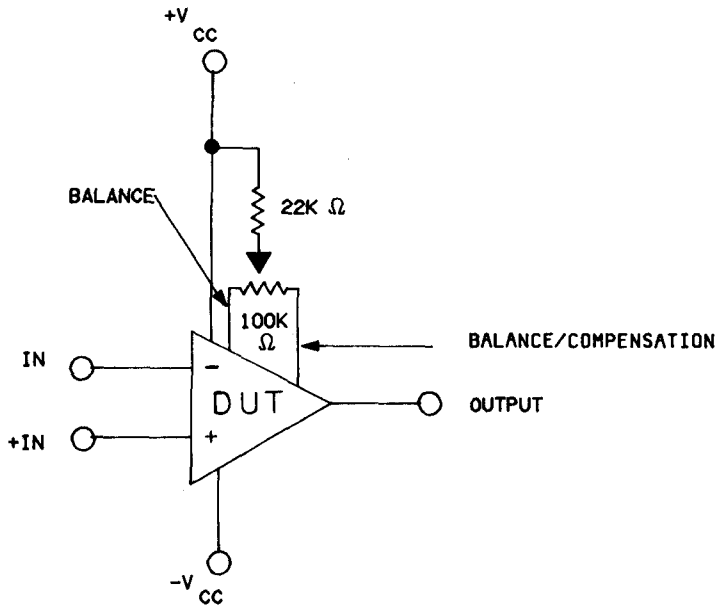
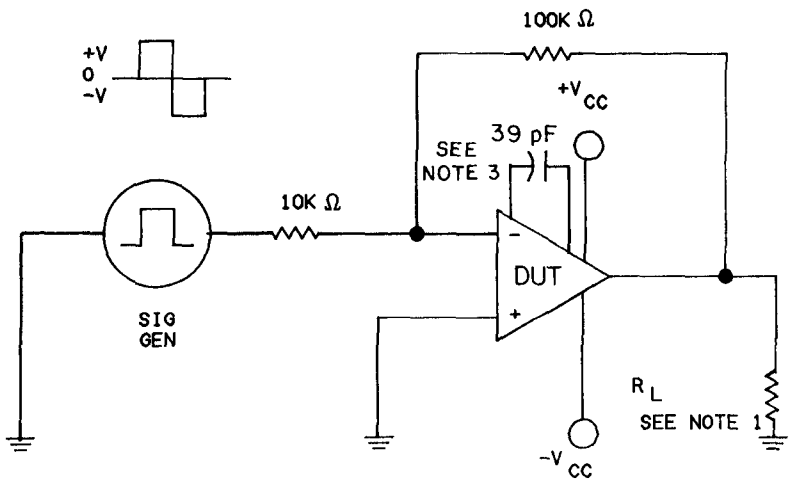


FIGURE 3. OFFSET NULL CIRCUIT



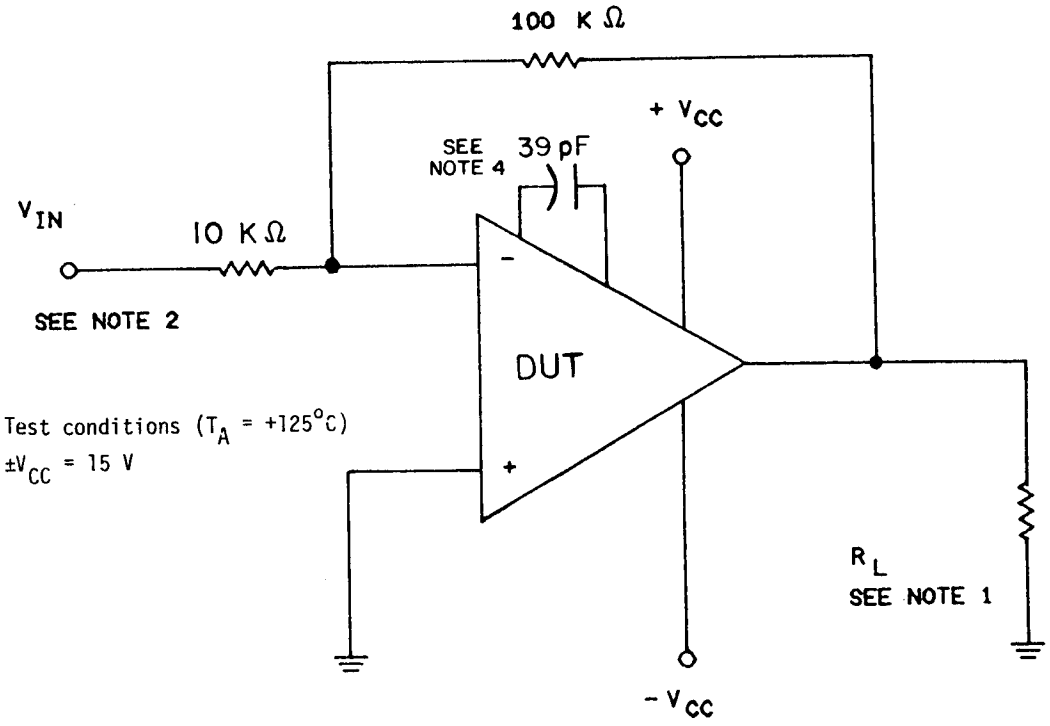
Test conditions (T_A=+125°C)
$\pm V_{CC}$ =15V
Sig.Gen. f=5Hz, 1V peak

NOTES:
1. Both halves as above dual amplifiers, R_L = 600Ω for device 01, 2 kΩ for amplifier in device 02.
2. All resistor tolerances are ±5%.
3. Device type 01 only.

FIGURE 4. Test circuit for burn-in and operating life tests.

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Device Types 01, 02

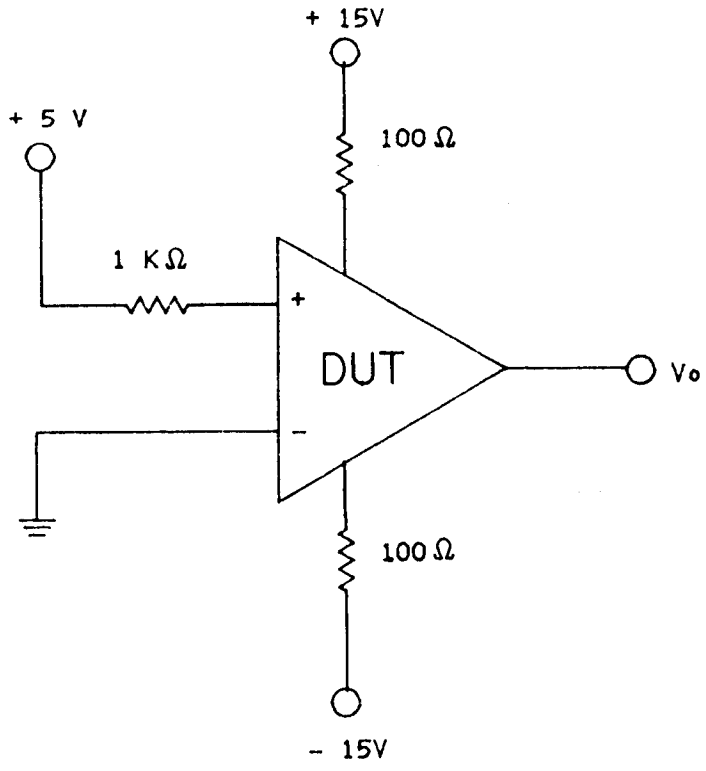


NOTES:
1. Both halves as above for dual amplifiers, R_L = 600Ω for device 01, 2 kΩ for each amplifier in device 02.
2. For device types 01 and 02, V_{IN} = 1 V dc.
3. All resistor tolerances are ±5%.
4. Device type 01 only.

FIGURE 5. Test circuit, burn-in (steady state power and reverse bias) and operating life test.

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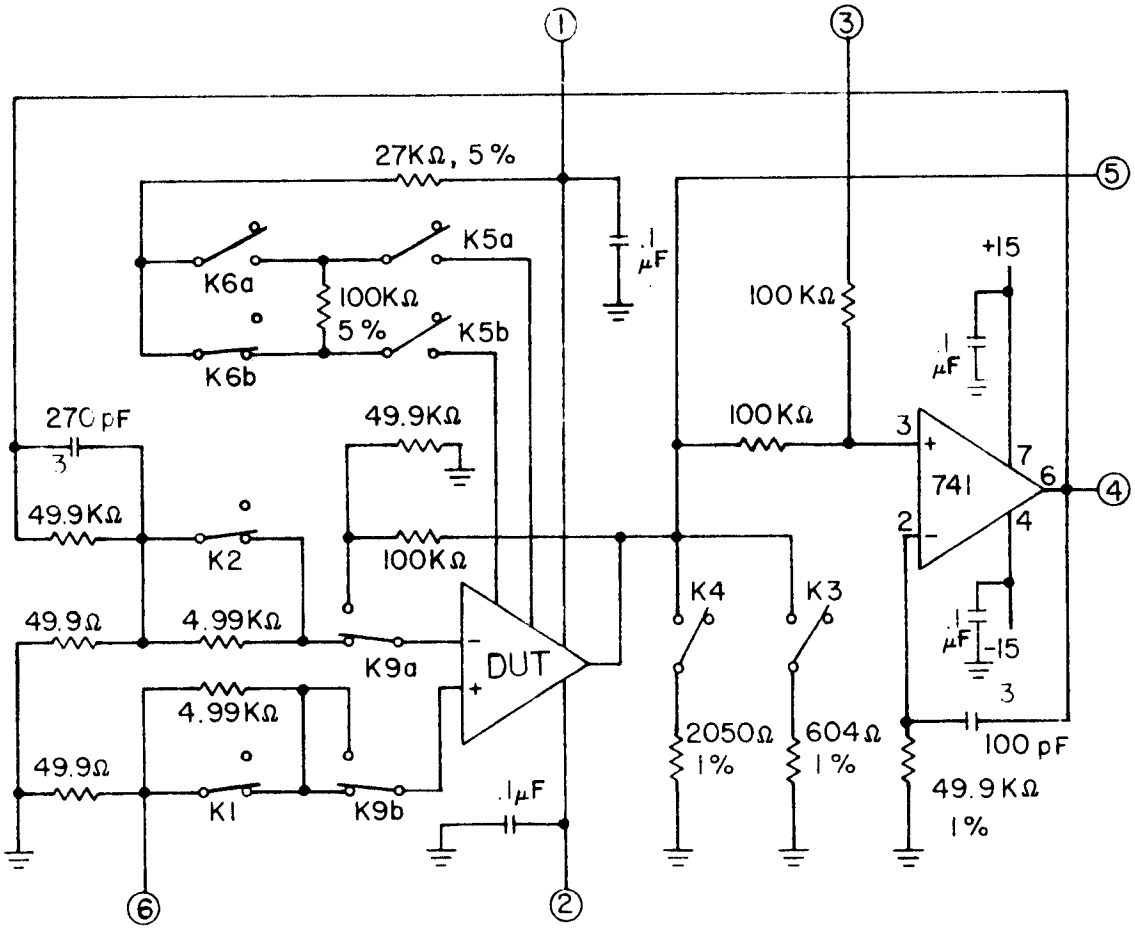
Device Types 01, 02



NOTE: If accelerated, high temperature test conditions are used, the device manufacturer shall ensure that at least 85% of the applied voltage is dropped across the device during test at temperature. The device is not considered fuctional under accelerated test conditions. For dual devices both halves are connected as above.

FIGURE 6. Accelerated burn-in and life test circuit.

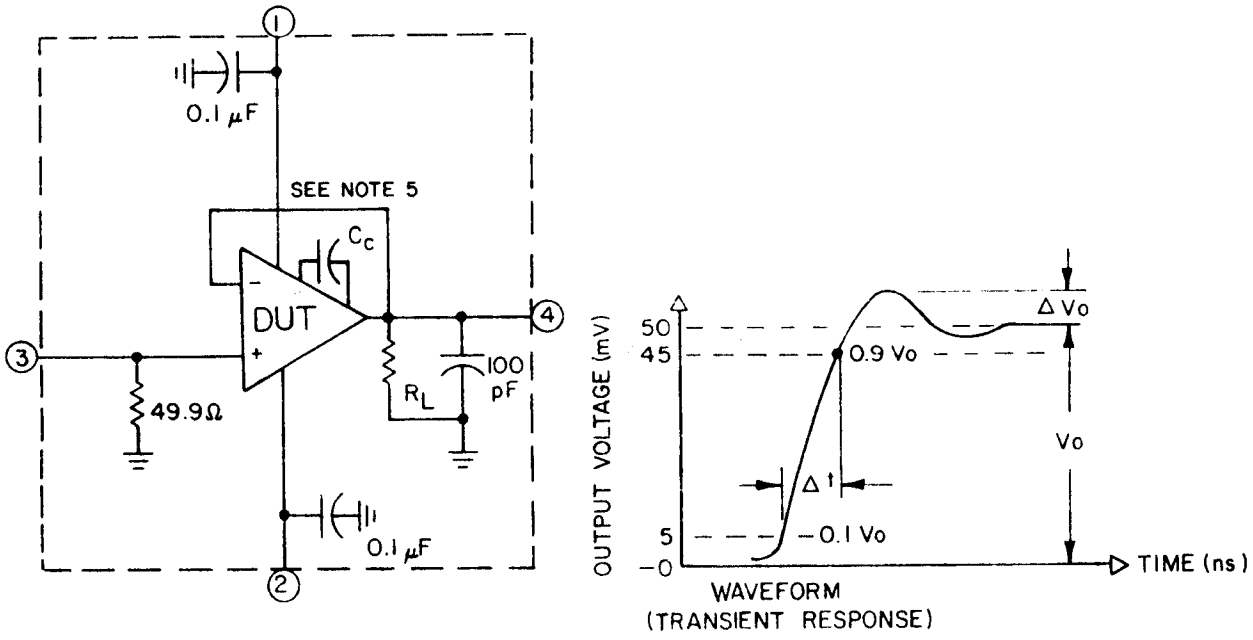
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NOTES:
1. All resistors are ±0.1% tolerance and all capacitors are ±10% unless specified otherwise.
2. Precautions shall be taken to prevent damage to the DUT during insertion into socket and change of relay state (i.e. disable voltage supplies, current limit $\pm V_{CC}$, etc.).
3. Compensation capacitors should be added as required for test circuit stability. Proper wiring precedures shall be followed to prevent unwanted coupling and oscillations, etc. Loop response and settling time shall be consistent with the test rate such that nay value has settled for at least 5 loop time constants before the value is measured.
4. Adequate settling time should be allowed such that each parameter has settled to within 5% of its final value.
5. All relays are shown in the normal de-energized state.
6. Saturation of the nulling amp is not allowed on tests where the pin 4 value is measured.
7. The load resistors 604Ω and 2050Ω yeild effective load resistances of 600Ω and 2000Ω respectively.
8. Any oscillation greater than 300 mV pk-pk is amplitude shall be cause for device failure.
9. Selection relays for dual devices are not shown.

FIGURE 7. Test circuit for static tests and slew rate.

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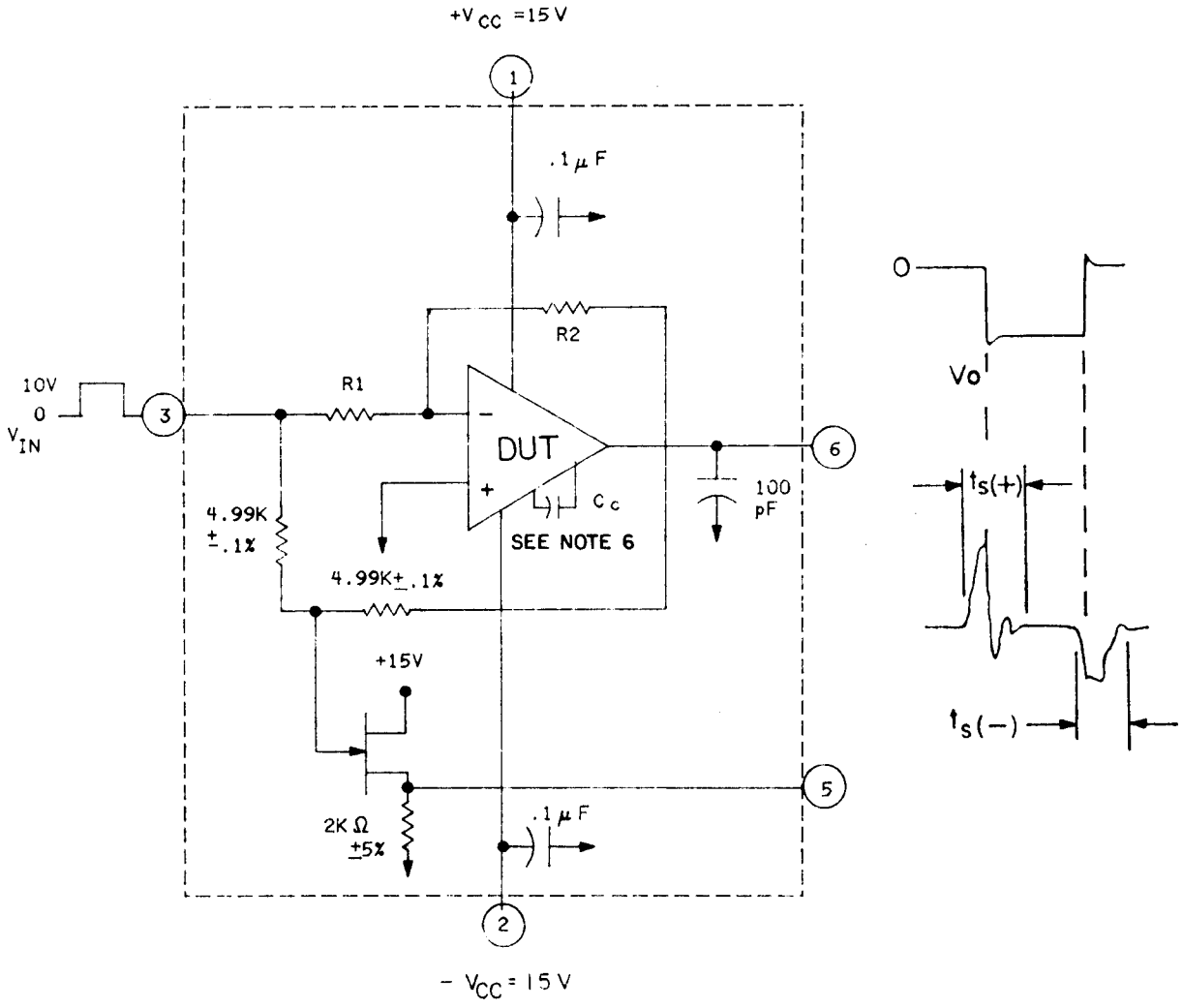


Parameter Symbol	Input pulse signal @ $t_r \leq 50$ ns	Equation
TR (t_r)	+50 mV	TR (t_r) = Δt
TR (o_s)	+50 mV	TR (o_s) = 100 (ΔVo/Vo)

NOTES:
1. Resistors are ±1% tolerance and capacitors are 10% tolerance.
2. Precaution should be taken to prevent damage to the DUT suring insertion into socket and in applying power.
3. R_L = 604Ω, C_C = 22 pF.
4. Selection circuitry for dual devices is not shown.
5. Device type 01 only.

FIGURE 8. Test circuit for transient response.

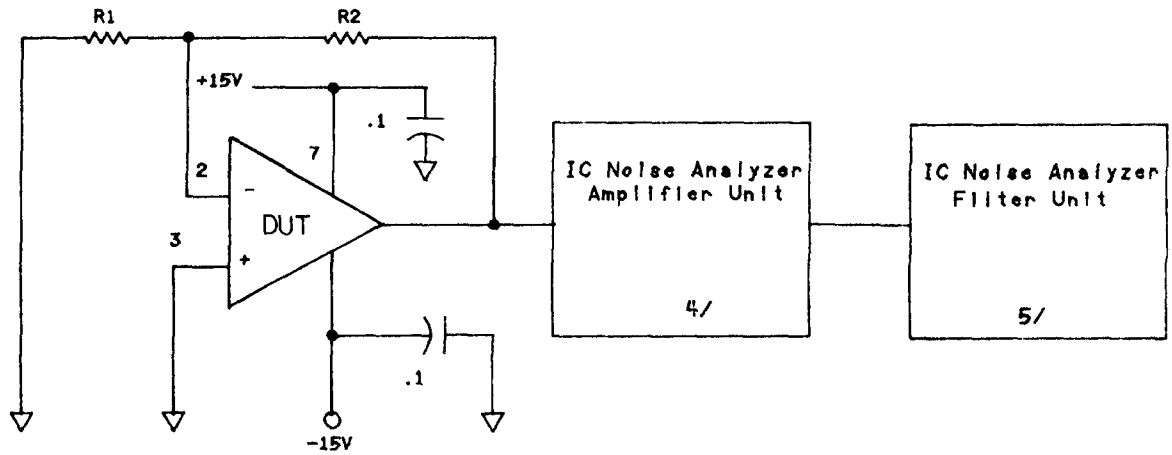
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NOTES:
1. Resistors are ±1% and capactiros are ±10% unless specified otherwise.
2. Precaution should be taken to prevent damage to the DUT during insertion into the socket and in applying power.
3. Selection circuitry for dual devices is not shown.
4. Settling time, as measured on pin 5, is defined as the interval of time during which the summing node is not nulled.
5. R1 = R2 = 604Ω, C_c = 22 pF.
6. Device type 01 only.

FIGURE 9. Test circuit for settling time.

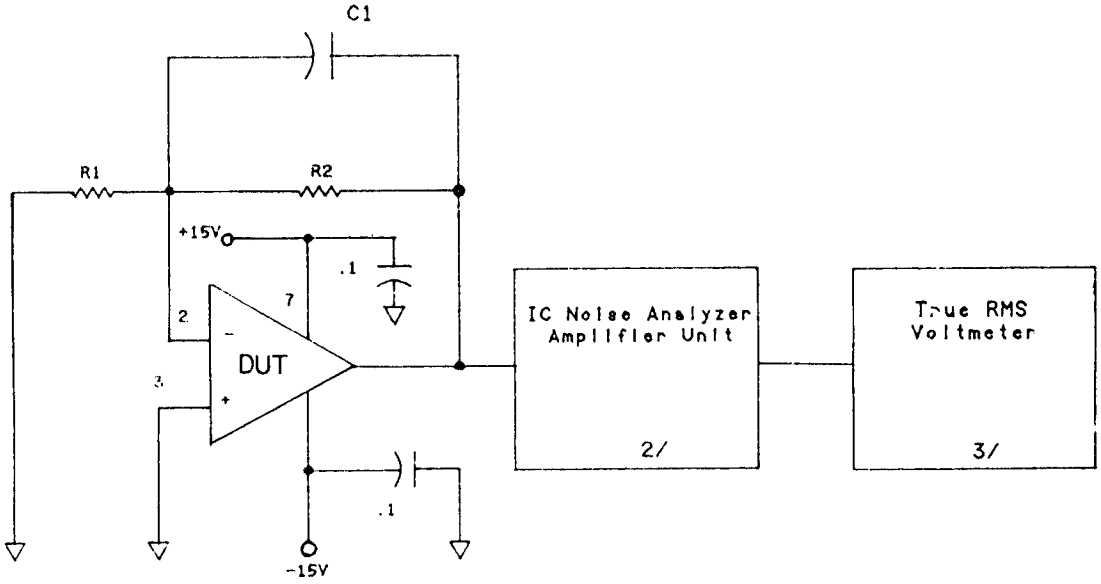
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NOTES:
1. Input Noise Voltage Density (En) Test: R1 = 50Ω, R2 = 10 kΩ.
2. Input Noise Current Density (In) Test: R1 = 105 kΩ, R2 = 2.0 MΩ.
3. All resistors are metal film and ±1% tolerance. Capacitors are in microfarads and are ±10% tolerance.
4. Quan-Tech Model 2283 or equivalent.
5. Quan-Tech Model 2181 or equivalent.

FIGURE 10. Noise density test circuit.

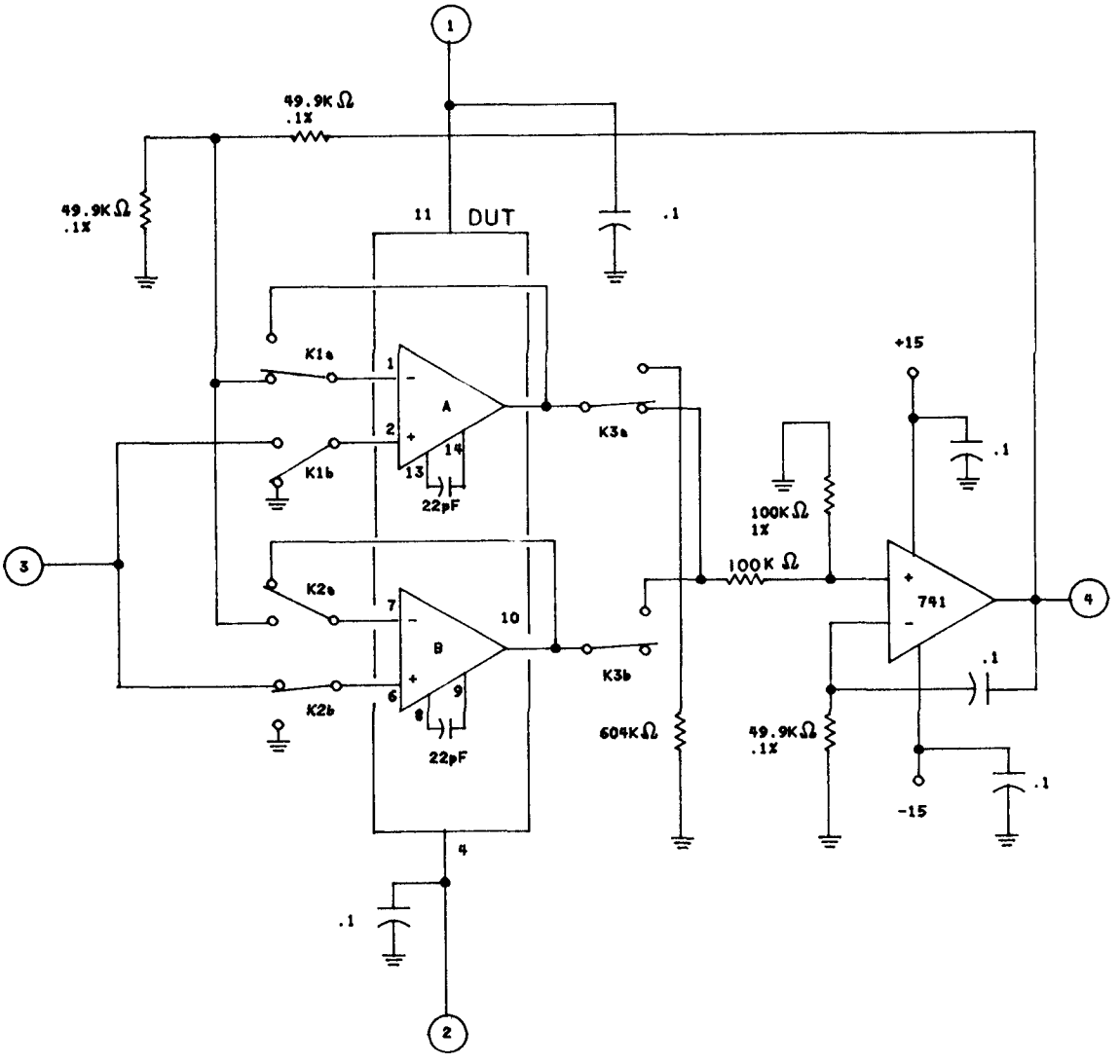
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NOTES:
1. All resistors are metal film and are ±1% tolerance. Capacitors are in microfarads and are ±10% tolerance.
2. Quan-Tech Model 2283 or equivalent. Bandwidth must be 10 Hz to 20 kHz minimum. Cain = 50.
3. Bandwidth must be 10 Hz to 20 kHz minimum.
4. Effective circuit noise banswidth is 10 kHz.
5. R1 = 50Ω, R2 = 10 kΩ, C1 = 2500 pF.

FIGURE 11. Broadband noise test circuit.

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NOTES:
1. Capacitors are ±10% tolerance.
2. All relay contacts are shown in the normal de-energized state.

FIGURE 12. Test circuit for channel separation.

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TABLE III. Group A inspection for all device types.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number 1	2	3	Energized relays	Measured pin No.	Value	Unit	Equation		Limits Min	Max	Unit
1 T_A = 25°C	V_{IO}	4001	1	6/	27	-3	-12	None	4	E1	V	V_{IO} = E1		-2	2	mV
			2		3	-27	12			E2		V_{IO} = E2				
			3		15	-15	0			E3		V_{IO} = E3				
			4		5	-5	0			E4		V_{IO} = E4				
	$+I_{IB}$		5		27	-3	-12	K1		E5		$+I_{IB}$ = 200 (E1-E5)		-800	800	nA
	$+I_{IB}$		6		15	-15	0	K1		E6		$+I_{IB}$ = 200 (E3-E6)				
	$+I_{IB}$		7		3	-27	12	K1		E7		$+I_{IB}$ = 200 (E2-E7)				
	$-I_{IB}$		8		27	-3	-12	K2		E8		$-I_{IB}$ = 200 (E8-E1)				
	$-I_{IB}$		9		15	-15	0	K2		E9		$-I_{IB}$ = 200 (E9-E2)				
	$-I_{IB}$		10		3	-27	12	K2		E10		$-I_{IB}$ = 200 (E10-E2)				
	I_{IO}		11	1/								I_{IO} = 200 (2E3-E6-E9)		-200	200	
	+PSRR	4003	12		20	-15	0	None	4	E11	V	+PSRR =20 Log(10000/(E11-E12))		85	-	dB
					10	-15	0			E12						
	-PSRR	4003	13		15	-20	0	None	4	E13	V	-PSRR =20 Log(10000/(E13-E14))		85	-	dB
					15	-10	0			E14						
	CMRR	4003	14	2/								CMRR = 20 Log(24000/(E1-E2))		80	-	dB
	$V_{IOADJ(+)}$		15		15	-15	0	K5	4	E15	V	$V_{IOADJ(+)}$ = E3-E15		4	-	mV
	$V_{IOADJ(-)}$		16		15	-15	0	K5,K6	4	E16	V	$V_{IOADJ(-)}$ = E3-E16		-	-4	mV
	$I_{OS(+)}$	3011	17	3/	12	-12	-10	None	5	I1	mA	$I_{OS(+)}$ = I1		-70	-	mA
	$I_{OS(-)}$	3011	18	3/	12	-12	-10	None	5	I2	mA	$I_{OS(-)}$ = I2		-	70	mA
	I_{CC}	4005	19		15	-15	0	None		I3	mA	I_{CC} = I3	01	-	6.5	mA
													02	-	11.0	mA

See footnotes at end of table.

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TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number 1	2	3	Energized relays	Measured pin No.	Value	Unit	Equation		Limits Min	Max	Unit
2 T_A = 125°C	V_{IO}	4001	20		27	-3	-12	None	4	E17	V	V_{IO} = E17		-3	3	mV
			21		3	-27	12			E18		V_{IO} = E18				
			22		15	-15	0			E19		V_{IO} = E19				
			23		5	-5	0			E20		V_{IO} = E20				
	dV_{IO}/dt		24	4/								$dV_{IO/dt}$ = 10(E19-E3)		-8	8	µV/°C
	$+I_{IB}$		25		27	-3	-12	K1	4	E21	V	$+I_{IB}$ = 200(E17-E21)		-1500	1500	nA
	$+I_{IB}$		26		15	-15	0	K1		E22		$+I_{IB}$ = 200(E19-E22)				
	$+I_{IB}$		27		3	-27	12	K1		E23		$+I_{IB}$ = 200(E18-E23)				
	$-I_{IB}$		28		27	-3	-12	K2		E24		$-I_{IB}$ = 200(E24-E17)				
	$-I_{IB}$		29		15	-15	0	K2		E25		$-I_{IB}$ = 200(E25-E19)				
	$-I_{IB}$		30		3	-27	12	K2		E26		$-I_{IB}$ = 200(E26-E18)				
	I_{IO}		31	1/								I_{IO} = 200(2E19-E22-E23)		-500	500	
	+PSRR	4003	32		20	-15	0	None	4	E27	V	+PSRR = 20Log(10000/(E27-E28))		85	-	dB
					10	-15	0			E28						
	-PSRR	4003	33		15	-20	0	None	4	E29	V	-PSRR = 20Log(10000/(E29-E30))		85	-	dB
					15	-10	0			E30						
	CMRR	4003	34	2/								CMRR = 20 Log(24000/(E17-E18))		80	-	dB
	$V_{IOADJ(+)}$		35	01 only	15	-15	0	K5	4	E31	V	$V_{IOADJ(+)}$ = E19-E31		4	-	mV
	$V_{IOADJ(-)}$		36	01 only	15	-15	0	K5,K6	4	E32	V	$V_{IOADJ(-)}$ = E19-E32		-	-4	mV
	$I_{OS(+)}$	3011	37	3/	12	-12	-10	None	5	I4	mA	$I_{OS(+)}$ = I4		-70	-	mA
	$I_{OS(-)}$	3011	38	3/	12	-12	10	None	5	I5	mA	$I_{OS(-)}$ = I5		-	70	mA
	I_{CC}	4005	39		15	-15	0	None		I6	mA	I_{CC} = I6	01	-	9	mA
													02	-	13	mA

See footnotes at end of table.

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TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number 1	2	3	Energized relays	Measured pin No.	Value	Unit	Equation		Limits Min	Max	Unit
3 T_A = -55°C	V_{IO}	4001	40		27	-3	-12	None	4	E33	V	V_{IO} = E33		-3	3	mV
			41		3	-27	12			E34		V_{IO} = E34				
			42		15	-15	0			E35		V_{IO} = E35				
			43		5	-5	0			E36		V_{IO} = E36				
	$dV_{IO/dt}$		44	4/								$dV_{IO/dt}$ = 12.5(E35-E3)		-8	8	µV/°C
	$+I_{IB}$		45		27	-3	-12	K1	4	E37	V	$+I_{IB}$ = 200(E33-E37)		-1500	1500	nA
	$+I_{IB}$		46		15	-15	0	K1		E38		$+I_{IB}$ = 200(E35-E38)				
	$+I_{IB}$		47		3	-27	12	K1		E39		$+I_{IB}$ = 200(E34-E39)				
	$-I_{IB}$		48		27	-3	-12	K2		E40		$-I_{IB}$ = 200(E40-E33)				
	$-I_{IB}$		49		15	-15	0	K2		E41		$-I_{IB}$ = 200(E41-E34)				
	$-I_{IB}$		50		3	-27	12	K2		E42		$-I_{IB}$ = 200(E42-E35)				
	I_{IO}		51	1/								I_{IO} = 200(2E35-E38-E41)		-500	500	
	+PSRR	4003	52		20	-15	0	None	4	E43	V	+PSRR = 20Log(10000/(E43-E44))		85	-	dB
					10	-15	0			E44						
	-PSRR	4003	53		15	-20	0	None	4	E45	V	-PSRR = 20Log(10000/(E45-E46))		85	-	dB
					15	-10	0			E46						
	CMRR	4003	54	2/								CMRR = 20 Log(24000/(E33-E34)		80	-	dB
	$V_{IOADJ(+)}$		55	01 only	15	-15	0	K5	4	E47	V	$V_{IOADJ(+)}$ = (E35-E47)		4	-	mV
	$V_{IOADJ(-)}$		56	01 only	15	-15	0	K5,K6	4	E48	V	$V_{IOADJ(-)}$ = (E35-E48)		-	-4	mV
	$I_{OS(+)}$	3011	57	3/	12	-12	-10	None	5	I7	mA	$I_{OS(+)}$ = I7		-70	-	mA
	$I_{OS(-)}$	3011	58	3/	12	-12	10	None	5	I8	mA	$I_{OS(-)}$ = I8		-	70	mA
	I_{CC}	4005	59		15	-15	0	None		I9	mA	I_{CC} = I9	01	-	9	mA
													02	-	13	mA

See footnotes at end of table.

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TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number 1	2	3	Energized relays	Measured pin No.	Value	Unit	Equation		Limits Min	Max	Unit
4 T_A = 25°C	$+V_{OP}$ $+V_{OP}$	4004	60 61	01 only	15	-15	-15	K3 K4	5	E49 E50	V	$+V_{OP}$ = E49 $+V_{OP}$ = E50	01 only	10 12	- -	V V
	$-V_{OP}$ $-V_{OP}$		62 63	01 only	15	-15	15	K3 K4		E51 E52		$-V_{OP}$ = E51 $-V_{OP}$ = E52	01 only	- -	-10 -12	V V
	$A_{VS(+)}$ $A_{VS(+)}$		64 65	01 only	15	-15	-10	K3 K4	4	E53 E54		$A_{VS(+)}$ = 10/(E3-E53) $A_{VS(+)}$ = 10/(E3-E54)		50	-	V/mV
	$A_{VS(-)}$ $A_{VS(-)}$		66 67	01 only	15	-15	10	K3 K4		E55 E56		$A_{VS(-)}$ = 10/(E55-E3) $A_{VS(-)}$ = 10/(E56-E3)				
	A_{VS}		68 69	01 only 01 only	5	-5	2 -2 2 -2	K3 K3 K4 K4		E57 E58 E59 E60		A_{VS} = 4/(E58-E57) A_{VS} = 4/(E60-E59)				
	EN		70 71	f=30 Hz f=1 kHz						E61 E62	nV nV	EN = $((E61)^2-.882)\cdot.5$ EN = $((E62)^2-.882)\cdot.5$		- -	11 5.5	nV/ Hz nV/ Hz
5 T_A = 125°C	$+V_{OP}$ $+V_{OP}$	4004	72 73	01 only	15	-15	-15	K3 K4	5	E63 E64	V	$+V_{OP}$ = E63 $+V_{OP}$ = E64	01 only	10 12	- -	V V
	$-V_{OP}$ $-V_{OP}$		74 75	01 only	15	-15	15	K3 K4		E65 E66		$-V_{OP}$ = E65 $-V_{OP}$ = E66	01 only	- -	-10 -12	V V
	$A_{VS(+)}$ $A_{VS(+)}$		76 77	01 only	15	-15	-10	K3 K4	4	E67 E68		$A_{VS(+)}$ = 10/(E19-E67) $A_{VS(+)}$ = 10/(E19-E68)		25	-	V/mV
	$A_{VS(-)}$ $A_{VS(-)}$		78 79	01 only	15	-15	10	K3 K4		E69 E70		$A_{VS(-)}$ = 10/(E69-E19) $A_{VS(-)}$ = 10/(E70-E19)				
	A_{VS}		80 81	01 only 01 only	5	-5	2 -2 2 -2	K3 K3 K4 K4		E71 E72 E73 E74		A_{VS} = 4/(E72-E71) A_{VS} = 4/(E74-E73)				

See footnotes at end of table.

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TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number 1	2	3	Energized relays	Measured pin No.	Value	Unit	Equation		Limits Min	Max	Unit
6 T_A = -55°C	$+V_{OP}$ $+V_{OP}$	4004	82 83	01 only	15	-15	-15	K3 K4	5	E75 E76	V	$+V_{OP}$ = E75 $+V_{OP}$ = E76	01 only	10 12	- -	V V
	$-V_{OP}$ $-V_{OP}$		84 85	01 only	15	-15	15	K3 K4		E77 E78		$-V_{OP}$ = E77 $-V_{OP}$ = E78	01 only	- -	-10 -12	V V
	$A_{VS(+)}$ $A_{VS(+)}$		86 87	01 only	15	-15	-10	K3 K4	4	E79 E80		$A_{VS(+)}$ = 10/(E35-E79) $A_{VS(+)}$ = 10/(E35-E80)		25	-	V/mV
	$A_{VS(-)}$ $A_{VS(-)}$		88 89	01 only	15	-15	10	K3 K4		E81 E82		$A_{VS(-)}$ = 10/(E81-E35) $A_{VS(-)}$ = 10/(E82-E35)				
	A_{VS}		90 91	01 only 01 only	5	-5	2 -2 2 -2	K3 K3 K4 K4		E83 E84 E85 E86		A_{VS} = 4/(E84-E83) A_{VS} = 4/(E86-E85)				
7 TA = 25°C	$T_{R(tr)}$		92	Fig. 8	15	-15	50 mV	None	4	E87	mV	$T_{R(tr)}$ = E87		-	40	ns
	$T_{R(os)}$		93	Fig. 8	15	-15	50 mV	None	4	E88	mV	$T_{R(os)}$ = E88		-	40	%
	EN EN		94 95	f=100Hz f=10kHz Fig. 10						E89 E90	nV	EN = $((E89)^2-.822)^{.5}$ EN = $((E90)^2-.822)^{.5}$		- -	9 5.0	nV/ Hz nV/ Hz
	IN		96 97 98 99	f=30Hz f=100Hz f=1kHz f=10kHz Fig. 10						E91 E92 E93 E94		IN = $(((91)^2-(61)^2-1644)/10000)^{.5}$ IN = $(((92)^2-(89)^2-1644)/10000)^{.5}$ IN = $(((93)^2-(62)^2-1644)/10000)^{.5}$ IN = $(((94)^2-(90)^2-1644)/10000)^{.5}$		- - - -	7 5 2 2	pA/ Hz pA/ Hz pA/ Hz pA/ Hz
	$S_{R(+)}$	4002	100		15	-15	0	K4,K9	5	E95	V	$S_{R(+)}$ = E95		10	-	V/µs
	$S_{R(-)}$	4002	101		15	-15	0	K4,K9	5	E96	V	$S_{R(-)}$ = E96		10	-	V/µs
	$N_{I(BB)}$		102	Fig. 11	15	-15	0	None	4	E97	µV	$N_{I(BB)}$ = E97/10		-	500	nVrms
	$C_{S(A)}$		103	Fig. 12 5/	15	-15	10 -10	None		E98 E99	V	$C_{S(A)}$ = 20 Log(20000/(E98-E99)		100	-	dB
	$C_{S(B)}$		104	Fig. 12 5/	15	-15	10 -10	K1,K2,K3		E100 E101	V	$C_{S(B)}$ = 20 Log(20000/(E100-E101)		100	-	dB

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TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin number			Energized relays	Measured pin			Equation	Limits		Unit
					1	2	3		No.	Value	Unit		Min	Max	
8 T_A = 125°C	$T_{R(tr)}$		105	Fig. 8	15	-15	50 mV	None	4	E102	mV	$T_{R(tr)}$ = E102	-	40	ns
	$T_{R(os)}$		106	Fig. 8	15	-15	50 mV	None	4	E103	mV	$T_{R(os)}$ = E103	-	40	%
	$S_{R(+)}$	4002	107		15	-15	0	K4,K9	5	E104	V	$S_{R(+)}$ = E104	10	-	V/µs
	$S_{R(-)}$	4002	108		15	-15	0	K4,K9	5	E105	V	$S_{R(-)}$ = E105	10	-	V/µs
8 T_A = -55°C	$T_{R(tr)}$		109	Fig. 8	15	-15	50 mV	None	4	E106	mV	$T_{R(tr)}$ = E106	-	40	ns
	$T_{R(os)}$		110	Fig. 8	15	-15	50 mV	None	4	E107	mV	$T_{R(os)}$ = E107	-	40	%
	$S_{R(+)}$	4002	111		15	-15	0	K4,K9	5	E108	V	$S_{R(+)}$ = E108	10	-	V/µs
	$S_{R(-)}$	4002	112		15	-15	0	K4,K9		E109	V	$S_{R(-)}$ = E109	10	-	V/µs
12 T_A = 25°C	$t_{s(+)}$		113	Fig. 9	15	-15	10	None		E110	mV	$t_{s(+)}$ = E110	-	4	µs
12 T_A = 25°C	$t_{s(-)}$		114	Fig. 9	15	-15	0	None		E111	mV	$t_{s(-)}$ = E111	-	4	µs

1/ I_{IO} is calculated using data from previous tests.

2/ CMRR is calculated using data from previous tests.

3/ $I_{OS(+)}$ and $I_{OS(-)}$ are measured with the output shorted to ground for less than 25 milliseconds.

4/ dv_{IO}/dt is calculated using data from previous tests. These read-and-record tests may be omitted except when subgroups 2 and 3 are being accomplished for group A sampling inspection and for groups C and D end point measurements.

5/ Device type 02 only.

6/ All tests apply to figure 7 unless otherwise specified.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

1. Class S devices - Test condition D using the circuit shown on figure 4.

2. Class B devices - Test condition D using the circuit shown on figure 4, or test condition C using the circuit shown on figure 5, or test condition F, using the circuit shown on figure 6.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). This screen test shall apply to class S devices only using the circuit shown on figure 5.

c. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA is specified as 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 9, 10, and 11 shall be omitted.

c. A special subgroup shall be added to group A inspection for class S devices only and shall consist of the tests, conditions, and limits of subgroup 12 as shown in table III herein.

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4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table IIb of method 5005 of MIL-STD-883, class B and as follows:

a. End-point electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 5. If the alternate burn-in conditions are used, the circuit shown on figure 5 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table IV herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B and C devices.

b. Steady state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

1. Test condition C, using the circuit shown on figure 5; or test conditions D, using the circuit shown on figure 4; or test condition F, using the circuit shown on figure 6.

2. T_A = 125°C minimum.

3. Test duration: 1000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Special subgroups shall be added to the group C inspection and shall consist of group A, subgroups 8 and 12, as specified in table III herein.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

TABLE IV. Group C end-point electrical parameters (T_A = 25°C)

Table III test no.	Test	Limit		Delta		Unit
		Min	Max	Min	Max	
103	V_{IO}	-2	2	-1	1	mV
106	$+I_{IB}$	-800	800	-400	400	nA
109	$-I_{IB}$	-800	800	-400	400	nA
111	I_{IO}	-200	200	-200	200	nA

T_A = 25°C, $\pm V_{CC}$ = ±15 V for all device types.

4.4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container shall be in accordance with MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Current values given are for conventional current and are positive when flowing into the referenced terminal.

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4.5.2 Burn-in and life test cooldown procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to an ambient temperature of 25°C ±3°C, and the electrical parameter end-point measurements shall then be performed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product to the procuring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variation in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type	Low noise operational amplifier description
01	5534A	Single, low noise, undercompensated
02	5532A	Dual, low noise, compensated

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Custodians:
Army - EK
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI, AR
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC,

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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